

## CURRICULUM VITAE

### Yuval Shmilovitz

CIRES postdoctoral fellow

University of Colorado Boulder, CO, USA

Email: [yuval.shmilovitz@colorado.edu](mailto:yuval.shmilovitz@colorado.edu)

Personal website: <https://yuvalsmilo.github.io/>

## EDUCATION

- 2014-2017 B.Sc. with high honor (Magna cum laude) at The Hebrew University of Jerusalem, Earth Sciences and History departments
- 2017-2019 M.Sc. at The Hebrew University of Jerusalem, the program of Hydrology and Water Resources. Advisors: Prof. Efrat Morin (HUJI), Prof. Yehuda Enzel, (HUJI), Prof. Itai Haviv (BGU). Thesis: *Linking storm-scale rainfall regime with arid slope morphology and evolution*
- 2019-2023 Ph.D. (direct track) at The Hebrew University of Jerusalem, the program of Hydrology and Water Resources. Advisors: Prof. Efrat Morin (HUJI), Prof. Yehuda Enzel, (HUJI), Prof. Itai Haviv (BGU). Thesis: *Climatic imprints on arid cliff-related hillslope evolution: linking hydrogeomorphic processes across time scales*
- 2024–present PostDoctoral fellowship at the University of Colorado Boulder, Cooperative Institute for Research in Environmental Sciences (CIRES). Host: the computational geomorphology group led by Prof. Gregory Tucker

## PROFESSIONAL EXPERIENCE, VISITS & TRAINING

- June 2018 – Aug. 2018 Research Assistant, excellent student's summer project, The Hebrew University of Jerusalem, Jerusalem, Israel. Supervisor: Prof. Dov Avigad
- Aug. 2018 Runoff Prediction in Ungauged Basins (PUB) summer school, Vienna University of Technology, Vienna, Austria
- Aug. 2020 Earth Surface Processes Institute (ESPI) summer school, University of Colorado Boulder, CO, USA
- May 2022 – July 2022 Visiting scientist, The Community Surface Dynamics Modeling System (CSDMS), University of Colorado Boulder, CO, USA
- 2024 – present PostDoctoral fellow, Cooperative Institute for Research in Environmental Sciences (CIRES), University of Colorado Boulder, CO, USA

## GRANTS AND FELLOWSHIPS

- 2021-2025 Collaborative Research, BSF-NSF (EAR-2100702/2100753) (\$239,902/\$239,902)\* *Deciphering the role of extreme rainstorms and hydroclimatic regime on arid escarpment retreat and sub-cliff slope evolution*
- 2022 Jewish National Fund (KKL-JNF) fellowship for climate change studies (\$13,500/\$13,500) (sole PI) *Soil erosion and characterization of extreme erosive events in arid areas*
- 2024-2026 CIRES Postdoctoral Grant, University of Colorado Boulder (\$150,000/\$150,000) (sole PI) *Understanding hydrogeomorphic impacts of rainstorm variability on dryland headwater catchments*
- 2024-2027 Jewish National Fund (KKL-JNF) Forestry Division (\$120,000)\*\* *Rainstorms and climate change impacts on soil erosion in the Northern Negev and effective preservation strategies*

\*Yuval Shmilovitz wrote the research proposal, but is not the official PI

*\*\*Yuval Shmilovitz is a research associate and collaborator who played a significant role in developing and writing the research proposal, but is not the official PI*

## AWARDS AND HONOURS

|                  |                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2014, 2015, 2016 | Dean's list for academic excellence (awarded 3 times), The Hebrew University of Jerusalem                                                |
| 2015             | Institute of Earth Sciences summer scholarship for excellent students, The Hebrew University of Jerusalem                                |
| 2018, 2019, 2020 | The Advanced School for Environmental Studies' scholarship for academic excellence (awarded 3 times), The Hebrew University of Jerusalem |
| 2021             | Outstanding paper by a PhD student, Israel Geographical Society                                                                          |
| 2022             | Student Research Award for Desert Research (Farouk El-Baz), Geological Society of America                                                |
| 2023             | Peretz Grader award for an excellent early-career scientist, Israel Geological Society                                                   |
| 2024             | Bentor Award (best Ph.D. dissertation), Institute of Earth Sciences, The Hebrew University of Jerusalem                                  |
| 2024             | CIRES Postdoctoral fellowship, University of Colorado Boulder                                                                            |

## TEACHING EXPERIENCE

|                   |                                    |                                                                                |
|-------------------|------------------------------------|--------------------------------------------------------------------------------|
| 2017 – 2022       | The Hebrew University of Jerusalem | The Dynamic Earth<br>(frontal teaching, B.Sc., 1st year)                       |
| 2019, 2021 – 2023 | The Hebrew University of Jerusalem | Geological Excursion in the Negev<br>(field teaching, B.Sc., 2st year)         |
| 2020              | The Hebrew University of Jerusalem | Northern Negev Field Trip<br>(field teaching, B.Sc., 2st year)                 |
| 2020 – 2022       | The Hebrew University of Jerusalem | Introduction to Geological Mapping<br>(field teaching, B.Sc., 2st year)        |
| 2020, 2021, 2023  | The Hebrew University of Jerusalem | Introduction to Geomorphology<br>(frontal & field teaching, B.Sc., 2/3st year) |
| 2021 – 2023       | The Hebrew University of Jerusalem | Summer Field Camp<br>(field teaching, B.Sc., 2st year)                         |

## OUTREACH, SUPERVISING AND MENTORING

|                      |                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 - 2022          | Bachelor project mentor, The Hebrew University of Jerusalem, <i>Rainfall spatiotemporal analysis in the central Negev using a convection-permitting climate model</i> |
| 2021 - 2022          | Bachelor project mentor, The Hebrew University of Jerusalem, <i>Optimization of stochastic rainfall generator for arid regions</i>                                    |
| 2021 - 2023          | Research project instructor (youth program), 'Kadim' educational center, Israel, <i>Geomorphology and hydrology across the Ramon cliffs and slopes</i>                |
| May 2025 & July 2024 | Project mentor, The Earth Surface Processes Institute (ESPI), CSDMS, University of Colorado Boulder                                                                   |

## REVIEWER SERVICE

- 2023 *Water Resources Research* (2)  
2024 *Geophysical Research Letters* (1), *Journal of hydrology* (2)  
2025 *NSF proposal ad hoc reviewer* (1), *Earth Surface Processes and Landforms* (1), *Journal of Geophysical Research: Earth Surface* (1)

## CONFERENCE SESSIONS CONVENED

- 2019 Co-convener: The 4th student seminar on innovations in hydrology and water resources, the Hebrew University of Jerusalem  
2022 Co-convener: Graduate student conference, Water Research in Israel, Technion, Haifa, Israel  
2022 Lead-convener (GM 10.2/CL 3.1/HS 13/NH 1): From hydro-climatology to hydro-geomorphology under extreme climatic events (at EGU annual meeting)  
2023 Lead-convener (GM 11.1): Geomorphic perspectives of environmental changes and extreme events in drylands and beyond (at EGU annual meeting)  
2024 Co-convener (GM 4.2): Hydrogeomorphic cascading processes and hazards impacted by environmental changes and extreme events (at EGU annual meeting)

## PRESS AND QUOTATIONS

- 2021 *Researchers are using ICESat-2 elevation data to create bathymetry maps of shallow lakes in the remote desert* NASA Earth Observatory  
2023 *Desert Landscape Evolution Controlled by Storm Intensity* AGU's EOS (Editor's highlights)  
2024 *Rainstorm Intensity Drives Desert Landscape Evolution* AGU's EOS (Editor's highlights)  
2025 *Dams failures intensified 2023 flood tragedy in Libya* CIRES Newsroom  
2025 *A Disaster foretold: The Failures That Led to the Deadliest Flood of the 21st Century* Yedioth-Net (ynet, in Hebrew)

## INVITED TALKS

- April 2021 **Shmilovitz, Y.**, *Frequency analysis of storm-scale soil erosion and characterization of extreme erosive events by linking process-based erosion model and a stochastic rainfall generator*, Volcani Institute - Agricultural Research Organization, Israel  
Nov. 2022 **Shmilovitz, Y.**, *Modeling long-term arid cliffs and sub-cliff slopes evolution under short-duration extreme rainstorm events*, CSDMS fall webinar  
June 2024 **Shmilovitz, Y.**, *On rainfall variability, erosion, and evolution of ephemeral landforms*, Colorado State University  
Feb. 2025 **Shmilovitz, Y.**, *Integration of high-resolution climate variability in the quantification of floods, erosion, and landscape evolution of ephemeral landforms*, Southwest Watershed Research Center, Tucson, AZ, USA  
May 2025 **Shmilovitz, Y.**, *On rainfall variability, erosion, and evolution of ephemeral landforms*, The Earth Surface Processes Institute, CSDMS, University of Colorado Boulder  
(Dec. 2025, upcoming) **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. *Climate-induced sediment budget transitions in dryland catchments: insights from landscape evolution numerical experiments across timescales*, AGU 2025 annual meeting

## PUBLICATIONS

- 12 Wei, H., Polyakov, V., Goodrich, D., Morin, E., Guertin, P. D., Assouline, S., Heilman P., Unkrich, C., **Shmilovitz, Y.** Marra, F., (2025). Modeling Runoff and Sediment Yield at the Event scale in Semiarid Watersheds. *International Soil and Water Conservation Research*. <https://doi.org/10.1016/j.iswcr.2025.07.001>.
- 11 Armon, M., **Shmilovitz, Y (equal contribution)**., & Dente, E. (2025). Anatomy of a foreseeable disaster: Lessons from the 2023 dam-breaching flood in Derna, Libya, *Science Advances*, *11*(13), eadu2865. <http://doi.org/10.1126/sciadv.adu2865>
- 10 **Shmilovitz, Y.**, Rossi, M. W., & Tucker, G. E. (2025). Multi-century erosion and landscape evolution of ephemeral catchments in response to sub-daily rainfall distribution changes, *Geophysical Research Letters*, *52*(5), e2024GL113179. <https://doi.org/10.1029/2024GL113179>
- 9 **Shmilovitz, Y.**, Tucker, G. E., Rossi, M. W., Morin, E., Armon, M., Pederson, J., Campforts, B., Haviv, I., & Enzel, Y. (2024). Impacts of rainstorm intensity and temporal pattern on caprock cliff persistence and hillslope morphology in drylands. *Journal of Geophysical Research: Earth Surface*, *129*(2), e2023JF007478. <https://doi.org/10.1029/2023JF007478>
- 8 **Shmilovitz, Y.**, Marra, F., Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Levi, Y., Khain, P., Rossi, Matthew W., Tucker, G., Pederson, J., & Haviv, I. (2023). The impact of extreme rainstorms on escarpment morphology in arid areas: Insights from the central Negev Desert. *Journal of Geophysical Research: Earth Surface*, *128*(10), e2023JF007093. <https://doi.org/10.1029/2023JF007093>
- 7 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Goodrich, D., Assouline, S., Morin, E. (2023). Assessing the controlling factors on watershed soil erosion during intense rainstorm events using radar rainfall and process-based modeling. *Catena*, *231*, 107282. <https://doi.org/10.1016/j.catena.2023.107282>
- 6 **Shmilovitz, Y.**, Shelef, E., Wieler, N., Zhang, H., Mushkin, A. (2023). Estimating the age of abandoned alluvial surfaces using morphologic dating of gully incision. *Journal of Geophysical Research: Earth Surface*, *128*(3), e2022JF006875. <https://doi.org/10.1029/2022JF006875>
- 5 **Shmilovitz, Y.**, Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Pederson, J., Haviv, I. (2023). Aspect-dependent bedrock weathering, cliff retreat, and cliff morphology in a hyperarid environment. *GSA Bulletin* 2022; *135* (7-8): 1955–1966. <https://doi.org/10.1130/B36442.1>
- 4 Wieler, N., Mushkin, A., Zhang, H., Sagy, A., Porat, N., **Shmilovitz, Y.**, Zhikun, R., Feipeng H., Piong, S., Jinrui, L., Shelef, E. (2022). Geomorphic Dating of Across-Fault Gully Incision Reveals Time-Invariant Late Quaternary Slip-Rates at the Eastern Termination of the Altyn Tagh Fault. *Geophysical Research Letters*, *49*(8), e2021GL096933. <https://doi.org/10.1029/2021GL096933>
- 3 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Nearing, M., Goodrich, D., Assouline, S., Morin, E. (2021). Frequency analysis of storm-scale soil erosion and characterization of extreme erosive events by linking the DWEPP model and a stochastic rainfall generator. *Science of The Total Environment*, *787*, 147609. <https://doi.org/10.1016/j.scitotenv.2021.147609>
- 2 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y. (2020). Linking frequency of rainstorms, runoff generation and sediment transport across hyperarid talus-pediment slopes. *Earth Surface Processes and Landforms*, *45*(7), 1644-1659. <https://doi.org/10.1002/esp.4836>
- 1 Armon, M., Dente, E., **Shmilovitz, Y.**, Mushkin, A., Cohen, T. J., Morin, E., Enzel, Y. (2020). Determining bathymetry of shallow and ephemeral desert lakes using satellite imagery and altimetry. *Geophysical Research Letters*, *47*(7), e2020GL087367. <https://doi.org/10.1029/2020GL087367>

## CONFERENCE PRESENTATIONS

- 15 **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. (2025). Ecohydrology and lithological controls on ephemeral catchments landscape evolution: insights from the High Plains (Colorado) using high-resolution climate projections. Poster, CSDMS 2025 annual meeting.
- 14 **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. (2025). Translating high-resolution climate change projections into erosion-vegetation feedbacks, sediment dynamics, and multi-century topographic evolution of dryland catchments. Oral presentation, EGU annual meeting.
- 13 **Shmilovitz, Y.**, Rossi, M., Tucker, G., (2024). Multi-century Sediment Yield and Topographic Evolution of Gullied Catchments: an Event-based Approach to Quantify the Effects of Changes in Rainfall Distribution, Oral presentation, AGU 2024 annual meeting
- 12 **Shmilovitz, Y.**, Rossi, M., Tucker, G., (2024). Gully development and soil erosion: An event-based approach using Landlab, Poster, CSDMS annual meeting
- 11 Dente, E., Armon, M., & **Shmilovitz, Y.** (2024). The September 2023 flood in Derna, Libya: an extreme weather event or man-made disaster? Oral presentation, EGU annual meeting
- 10 **Shmilovitz, Y.**, Rossi, M., Tucker, G., Campforts, B., Pederson, J., Morin, E., Armon, M., Enzel, Y., Haviv, I. (2023). Simulating dryland cliffs evolution in response to extreme rainstorms, Poster, EGU annual meeting
- 9 **Shmilovitz, Y.**, Shelef, E., Wieler, N., Huiping, Z., Mushkin, A., (2022). Estimating the age of abandoned alluvial surfaces using morphologic dating of gully incision, Oral presentation, GSA Annual Meeting 2022
- 8 **Shmilovitz, Y.**, Marra, F., Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Levi, Y., Khain, P., Haviv, I., (2022). The signature of extreme rainstorms properties on cliff morphology in arid areas, Pico, EGU annual meeting
- 7 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Nearing, M., Goodrich, D., Assouline, S., Morin, E., (2021). Frequency analysis of storm-scale soil erosion and characterization of extreme erosive events by linking process-based erosion model and a stochastic rainfall generator, Oral presentation, Eyal annual meeting, Zichron Yaakov, Israel
- 6 **Shmilovitz, Y.**, Mushkin, A., Wieler, N., Zhang, H., Shelef, E., (2021). Morphologic dating of gully incision on abandoned alluvial surfaces using a landscape evolution model, Oral presentation, The Israel Geological Society Annual Conference, Yerocham, Israel
- 5 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Nearing, M., Goodrich, D., Assouline, S., Morin, E., (2020). The role of rainstorm properties on crop-land soil erosion: coupling event-scale modeling with a stochastic rainfall generator for estimating erosion risks, Poster, AGU annual meeting
- 4 **Shmilovitz, Y.**, Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Haviv, I., (2020). Aspect-dependent escarpment retreat, sediment mobilization and slope evolution under hyper-arid conditions., Poster., GSA annual meeting
- 3 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y., (2020). Linking frequency of rainstorms, runoff generation and sediment transport across hyperarid talus-pediment slopes, Pico, EGU annual meeting
- 2 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y., (2019). Linking frequency of rainstorms, runoff generation, and sediment transport: Implications for arid hillslopes evolution and reorganization, Poster, EGU Leonardo Conference, Esch-sur-Alzetter, Luxembourg
- 1 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y., (2019). Linking hydro-metrology and geomorphology across hyper-arid talus-pediment slopes: frequency of runoff generating storms, sheetwash efficiency and form-processes relationships, Oral presentation, The Israel Geological Society annual conference, Kfar-Blum, Israel